

ARISTO EDUCATIONAL PRESS
HONG KONG DIPLOMA OF SECONDARY
EDUCATION EXAMINATION

(Mock Test 33)

Suggested Solution

Paper 1

- | | | |
|-------|-------|-------|
| 1. A | 11. B | 21. A |
| 2. C | 12. C | 22. C |
| 3. B | 13. B | 23. B |
| 4. B | 14. C | 24. D |
| 5. D | 15. C | 25. B |
| 6. B | 16. C | 26. C |
| 7. D | 17. C | 27. A |
| 8. D | 18. D | 28. A |
| 9. A | 19. A | 29. B |
| 10. A | 20. D | 30. D |

Explanation

1. Answer: A

The time and money spent by Carmen on studying nursing is a cost that has been paid in the past and cannot be recovered, so it is not the opportunity cost of the choice and it does not affect current decision-making.

Option B is incorrect. Shorter working hours for teaching piano full-time is related to the benefits of changing her occupation.

Option C is incorrect. Opportunity for promotion in a public hospital is related to the cost of changing her occupation.

Option D is incorrect. The income from teaching piano is related to the benefits of changing her occupation.

2. Answer: C

The full cost of taking a bus to Tsim Sha Tsui includes the bus fare (money cost) and time cost. Traffic congestion will lead to a longer bus journey, thus increasing the time cost. The full cost will increase.

3. Answer: B

(1) is incorrect. In a market economy, people are not necessarily wealthier than in a planned economy.

(3) is incorrect. Public goods are not necessarily provided by the government. Private firms can also provide public goods. The quantity of public goods has nothing to do with the type of economic system. The quantity of public goods may not decrease in a market economy.

4. Answer: B

(2) is incorrect. Buying debentures can earn a fixed interest rate, while buying shares can earn a variable dividend rate and capital gain. Thus, buying shares may earn him a higher return than buying debentures.

(4) is incorrect. Debenture holders have no voting rights in the company's decision-making.

5. Answer: D

A private limited company cannot issue shares or bonds to the public to raise capital, while a public limited company can.

Option A is incorrect. Shareholders of both a private limited company and a public limited company may not directly participate in the management of the firm.

Option B is incorrect. Both a private limited company and a public limited company are legal entities.

Option C is incorrect. A public limited company is required to disclose its accounting information to the public regularly while a private limited company is not required to disclose its accounting information to the public.

6. Answer: B

(1) is correct. The decorations are goods used to produce a final service (photo opportunities) desired by consumers, so they are producer goods.

(4) is correct. The decorations are non-rival and non-excludable in consumption. Their consumption is concurrent in nature and therefore they are public goods.

(3) is incorrect. Since the quantity of decorations available is not sufficient to satisfy all human wants and people would prefer more of them, they are economic goods instead of free goods.

7. Answer: D

(3) is correct. Having the same tutor teaches classes of the same grade level results in a more standardised class than having different tutors teaching the same grade level.

8. Answer: D

Average labour productivity of firm A = $850 \text{ units} / (10 \times 8.5) \text{ man-hours} = 10 \text{ units/man-hour}$

Average labour productivity of firm B = $1,000 \text{ units} / (12 \times 8) \text{ man-hours} = 10.42 \text{ units/man-hour}$

Compared with firm B, firm A has a lower average labour productivity and a higher percentage of output with quality up to standard. This means that the workers of firm A may be paid by time rate and do not need to rush for more output at the expense of quality. On the other hand, the average labour productivity of firm B is higher, but the percentage of output with quality up to standard is lower. This means that the workers of firm B may be paid by piece rate and they rush for more output at the expense of quality.

9. Answer: A

The provision of more vocational training courses can improve people's skills, thereby increasing average labour productivity and increasing people's ability and willingness to change from one occupation to another.

Option B is incorrect. The increase in the quota of the skilled immigration scheme will increase average labour productivity, but occupational mobility of labour will not be affected.

Option C is incorrect. Setting up a website to provide information on job vacancies will increase occupational mobility of labour, but average labour productivity will not be affected.

Option D is incorrect. More professional groups tightening their licensing requirements will make it more difficult for people to change occupation, which will reduce occupational mobility of labour, while the effect on average labour productivity is uncertain.

10. Answer: A

(1) is correct. Even if the business of insurance companies is not good, insurance agents can still receive a basic salary, which makes their income more stable.

(2) is incorrect. This is an advantage of the arrangement from the viewpoint of insurance companies, but not an advantage from the viewpoint of insurance agents.

(3) is incorrect. Insurance agents can increase their income by improving performance under both basic salary plus commission and commission only.

11. Answer: B

Machinery (units)	Labour (units)	Total product (units)	Marginal product (units)
1	10	5	5
2	10	15	10
3	10	28	13
4	10	Q_1	X
5	10	Q_2	Y

Option B is correct. When $Q_1 = 43$ and $Q_2 = 60$, $X = 15$ and $Y = 17$. The data cannot illustrate the law of diminishing marginal returns because the marginal product continues to increase.

Option A is incorrect. When $Q_1 = 40$ and $Q_2 = 50$, $X = 12$ and $Y = 10$. Since the marginal product first increases and then decreases, the data illustrates the law of diminishing marginal returns.

Option C is incorrect. When $Q_1 = 45$ and $Q_2 = 42$, $X = 17$ and $Y = -3$. Since the marginal product first increases and then decreases, the data illustrates the law of diminishing marginal returns.

Option D is incorrect. When $Q_1 = 45$ and $Q_2 = 60$, $X = 17$ and $Y = 15$. Since the marginal product first increases and then decreases, the data illustrates the law of diminishing marginal returns.

12. Answer: C

The fitness center group may suffer internal diseconomies of scale because expanding its production scale may increase its average cost. As consumers did not respond satisfactorily to the firm's additional promotion, the increase in the number of customers may not be able to cover the increase in marketing expenses, resulting in an increase in average cost.

13. Answer: B

When the demand for second-hand designer handbags increases, the market price will increase. The marginal revenue will increase and the profit-maximising output will increase.

Option A is incorrect. When the monthly rental payment of the store decreases, the fixed cost of selling second-hand designer handbags will decrease and the profit-maximising output will not be affected.

Option C is incorrect. When the wage rate of workers who repair designer handbags increases, the marginal cost of selling second-hand designer handbags will increase and the profit-maximising output will decrease.

Option D is incorrect. When acquiring second-hand designer handbags becomes more difficult, the marginal cost of selling second-hand designer handbags will increase and the profit-maximising output will decrease.

14. Answer: C

When the average cost per unit of output falls by \$2:

Output (units)	Average cost (\$)	Total cost (\$)	Marginal cost (\$)
1	8	8	6
2	9	18	10
3	10	30	12
4	11	44	14
5	12	60	16

The market price is \$12 and the profit-maximising output is 3 units ($MC = MR = \text{Price}$).

Maximum profit = $\$12 \times 3 - \$30 = \$6$

Option A is incorrect. There is a fixed cost, which shows that the firm produces in the short run, and diseconomies of scale do not apply to short-run production.

Option B is incorrect. The profit-maximising output will increase from 2 units to 3 units.

Option D is incorrect. If the firm chooses to produce 1 unit, the total cost (\$8) will be lower than \$30.

15. Answer: C

When deriving the market supply curve for a good, economists assume all the factors affecting a producer's quantity supplied other than its price hold constant, to study the effect of changes in prices on quantities supplied.

16. Answer: C

(2) is correct. After the increase in price, the quantity transacted will decrease and consumer surplus will decrease.

(3) is correct. After adding \$1 to the prices of all products, the relative price of higher-priced meals (in terms of lower-priced meals) will fall. According to the law of demand, people will buy more higher-priced meals relative to lower-priced meals. The proportion of higher-priced meals sold to lower-priced meals sold will increase.

(1) is incorrect. The quantity transacted will decrease after the increase in price. The change in total revenue is uncertain, depending on the price elasticity of demand for restaurant services.

17. Answer: C

If good X and good Y are in joint demand, when the price of good X increases, the demand for good Y will decrease. Both the price and quantity transacted will decrease. The producer surplus in the market of good Y will decrease.

Option A is incorrect. If good X and good Y are in joint supply, when the price of good X increases, the supply of good Y will increase. Its price will decrease and the quantity transacted will increase.

Option B is incorrect. If good X and good Y are in competitive supply, when the price of good X increases, the supply of good Y will decrease. Its price will increase and the quantity transacted will decrease. Total revenue from the sale of good Y may increase, decrease or remain unchanged, depending on the price elasticity of demand for good Y.

Option D is incorrect. If good X and good Y are in competitive demand, when the price of good X increases, the demand for good Y will increase. Both its price and quantity transacted will increase. The producer surplus in the market of good Y will increase.

18. Answer: D

When the government abolishes restrictions on importing foreign workers, more foreign workers can come to work, and the supply of construction workers will increase.

Option A is incorrect. When there are more jobs opportunities in other industries, more construction workers may seek employment in other industries, and the supply of construction workers may decrease.

Option B is incorrect. An increase in the average wage rate for construction workers will lead to an increase in the quantity supplied of construction workers. The supply of construction workers will not be affected.

Option C is incorrect. When the government announces the construction of more public housing, the demand for construction workers will increase.

19. Answer: A

Given that cruise services are normal goods, when people's income increases, the demand for cruise services will increase. The demand curve will shift to the right from D_0 to D_1 .

When the maintenance cost of cruise ships increases sharply, the cost of providing cruise services will increase. The supply of cruise services will decrease and the supply curve will shift to the left from S_0 to S_2 .

20. Answer: D

When the price increases, Miss Wong's quantity demanded remains unchanged (4 units), showing that her demand for bread is perfectly inelastic and the value of the price elasticity of demand coefficient is 0.

When the price increases, Mr. Li's total expenditure decreases, showing that his demand for bread is elastic and the value of the price elasticity of demand coefficient is greater than 1 and smaller than infinity.

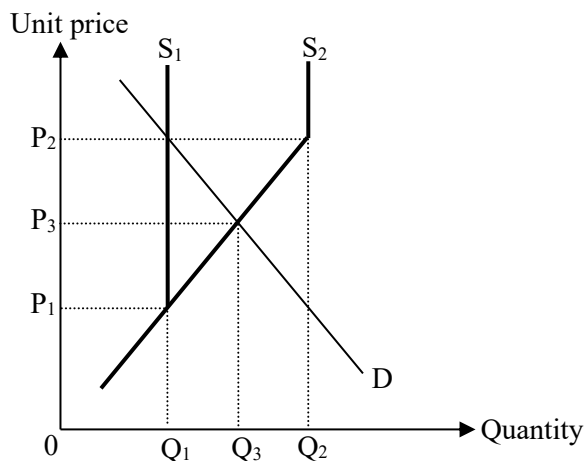
21. Answer: A

Given that the demand for good X is a derived demand for good Y, when the demand for good Y decreases, the demand for good X will also decrease. The price and quantity transacted of good X will decrease.

$$\text{Percentage decrease in price} = (\$8 - \$10) / [(\$10 + \$8) / 2] \times 100\% = 22.22\%$$

Since the percentage change in quantity supplied (30%) is higher than the percentage change in price (22.22%), the supply of good X is elastic.

22. Answer: C



(2) is correct. When the quota limit is Q_2 , the quota is ineffective. The quantity demanded and quantity supplied will both be Q_3 and the market of good X will be in equilibrium.

(3) is correct. When the quota limit increases from Q_1 to Q_2 , the quota will become ineffective. The quantity transacted will increase from Q_1 to Q_3 and the marginal benefit of the last unit of good X will decrease from P_2 to P_3 .

(1) is incorrect. When the quota limit is Q_1 , the price is P_2 and the total expenditure is $(P_2 \times Q_1)$. When the quota limit is Q_2 , the quota is ineffective. The price is P_3 and the total expenditure is $(P_3 \times Q_3)$.

23. Answer: B

Original price = \$10

Original quantity transacted = 25 units

Unit price (\$)	Quantity demanded (units)	Quantity supplied (units)	Quantity supplied after tax (units)
6	35	15	-
8	30	20	-
10	25	25	15
12	20	30	20
14	15	35	25
16	10	40	30

New quantity transacted = 20 units

Total tax revenue = $\$4 \times 20 \text{ units} = \80

The new price is \$12. Since consumers' tax burden (\$2 per unit) equals producers' tax burden (\$2 per unit), the price elasticity of demand is equal to the price elasticity of supply.

24. Answer: D

Option C is incorrect. Firms under oligopoly produce heterogeneous products, but this is not sufficient to support the answer because firms of other imperfectly competitive markets (e.g. monopolistic competition) also produce heterogeneous products.

25. Answer: B

Prices serve the function of rationing or distributing goods to users with the highest value for the goods.

Option A is incorrect. Although the case illustrates the rationing function of price, the explanation that quantity demanded will be raised with a lower price does not match the definition of the rationing function of price.

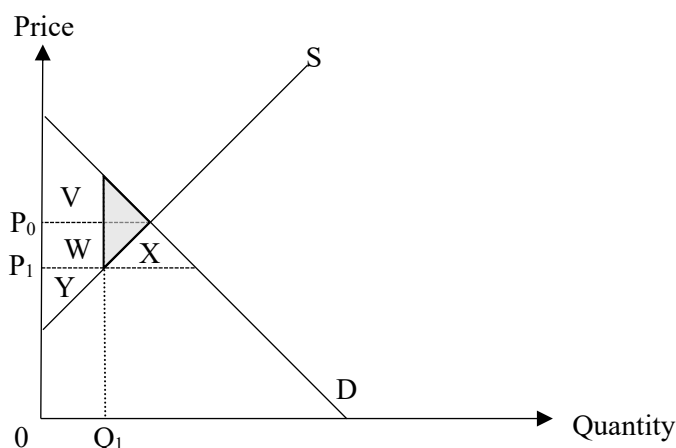
26. Answer: C

When the government sets the price ceiling on good X at P_1 , the quantity transacted is Q_1 , and the producer surplus is equal to area Y.

Option A is incorrect. The consumer surplus is equal to area $(V + W)$ minus the deadweight loss (the grey part in the figure below).

Option B is incorrect. The total social surplus is equal to area $(V + W + Y)$ minus the deadweight loss (the grey part in the figure below).

Option D is incorrect. The deadweight loss is the grey part in the figure below, which is not equal to area X.



27. Answer: A

(1) is correct. Imposing the tax will lead to a decrease in the supply of plastic tableware. The equilibrium price will increase and the quantity transacted will decrease. Consumer surplus will decrease.

(2) is correct. Imposing the tax will cause the equilibrium price of plastic tableware to increase and the quantity transacted to decrease. If consumers' demand for plastic tableware is inelastic, the percentage increase in price will be greater than the percentage decrease in quantity demanded, and consumers' total expenditure will increase.

(3) is incorrect. After imposing the tax, the equilibrium price of plastic tableware will increase. Wooden tableware and plastic tableware are substitutes. When the price of plastic tableware increases, the demand for wooden tableware will increase and the total social surplus will increase.

28. Answer: A

(1) and (2) are correct. Since they increase the revenue of the MTR Corporation, they are private benefits to the company.

(3) is incorrect. The opening of the shopping mall has benefited the owners of nearby private residential properties, but they do not give any compensation, so it is an external benefit.

29. Answer: B

Country A's Lorenz curve in 2023 is closer to the line of perfect income equality than the Lorenz curve in 2020, showing that income inequality has improved in 2023. A progressive tax system can help improve income inequality.

Option A is incorrect. Reducing the transfer payments to the poor will worsen income inequality.

Option C is incorrect. After reducing the basic allowance of the salaries tax, those low-income people who did not have to pay tax before may have to pay tax. The post-tax post-social transfer income of low-income people will decrease, worsening income inequality.

Option D is incorrect. The Lorenz curve shows the income distribution of the economy and does not reflect the income of households.

30. Answer: D

(1) and (2) are correct. Persons with disabilities are usually low-income people. Providing subsidies to employers who hire persons with disabilities can improve their chances of being hired and help increase their income.

(3) is correct. Under the scheme, employers will provide persons with disabilities with appropriate training. This can help raise their labour productivity.

Paper 2

Section A (30 marks)

1. (a) No. (1 mark)
Members-only videos are excludable in consumption. The video platform can exclude users who do not pay from viewing these videos. (2 marks)
- (b) Possible reasons:
- different quality of videos
- different production costs
- any reasonable answer
(Mark the **FIRST** point only) } 2@, max: 2 marks
- (c) Complex division of labour, (1 mark)
because each worker engages in a different stage of video production. (1 mark)
A reason why division of labour can increase labour productivity:
- choosing the best people to do the job
- practice makes perfect
- saving time on training
- any reasonable answer
(Mark the **FIRST** point only) } 1@, max: 1 mark
2. (a) Private limited company. (1 mark)
- (b) An advantage:
- A private limited company has an independent legal status.
- A private limited company has lasting continuity.
- A private limited company has wider sources of capital.
- Any reasonable answer
(Mark the **FIRST** point only) } 1@, max: 1 mark
- A disadvantage:
- A private limited company is subject to a higher profits tax rate.
- There are more complicated set-up procedures for a private limited company.
- Any reasonable answer
(Mark the **FIRST** point only) } 1@, max: 1 mark

3. (a) Capital. (1 mark)

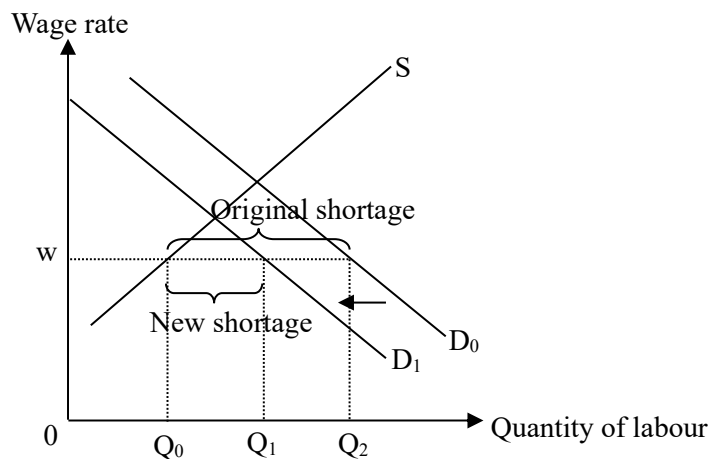
(b) (i) Verbal elaboration:

There is a shortage in the labour market. Robots and labour are substitutes. When robots are introduced, the demand for labour will decrease. Supposing the wage rate for labour is not adjusted, the decrease in demand will reduce the shortage of labour. (3 marks)

Illustrate in the diagram:

- leftward shift of the demand curve (1 mark)

- correct positions of the original and the new shortages (a decrease in the shortage) (1 mark)



(ii) Positive statement, because it can be refutable by facts. (2 marks)

4. For:

After increasing the fees, the fees charged for the recreational and sports facilities will be closer to the equilibrium price and the gap between marginal benefit and marginal cost will be narrowed. Economic efficiency will be improved. (2 marks)

OR

Any reasonable answer (2 marks)

Against:

After increasing the fees, low-income people may not be able to afford the fees and are therefore deprived of the opportunity to use recreational and sports facilities to practise the sports or maintain their health, thus harming their ability to earn income. High-income people will not be affected. This does not help equalise opportunities and equity will be worsened. (2 marks)

OR

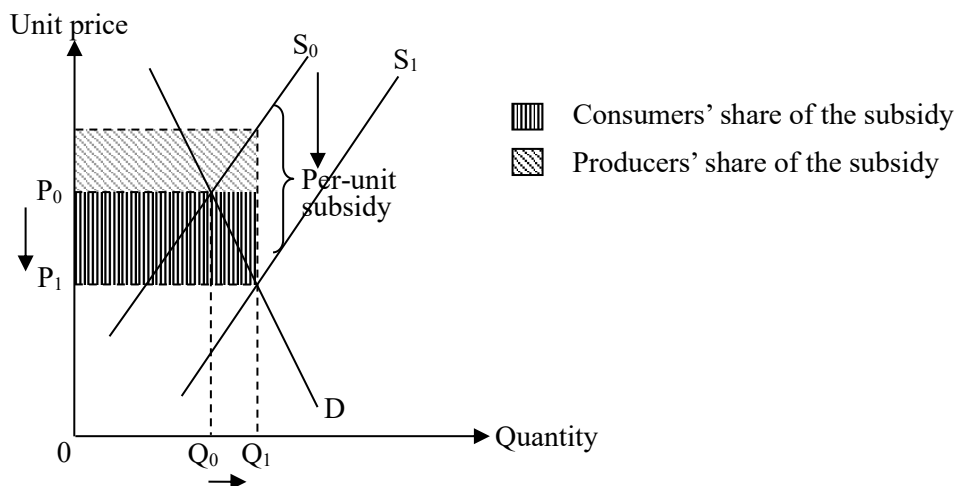
Any reasonable answer (2 marks)

5. Verbal elaboration:

The condition is that the supply of private medical services is more elastic than the demand. (2 marks)

Illustrate in the diagram:

- downward shift of the supply curve (1 mark)
- correct position of the new price (P_1) (1 mark)
- correct position of the new quantity transacted (Q_1) (1 mark)
- correct position of consumers' share of the subsidy and producers' share of the subsidy (1 mark)
- consumers' share of the subsidy > producers' share of the subsidy (1 mark)



Section B (40 marks)

6. (a) Vertical forward expansion. (1 mark)

An advantage:

- to ensure a market outlet for the products of the bee farm
- to take advantage of a brand name
- any reasonable answer

1@, max: 1 mark

(Mark the **FIRST** point only)

(b) From the viewpoint of Mr. Cheung:

- The cost of measuring the clerk's output is lower because wages are paid on a time basis.
- The cost of monitoring output quality is lower because the clerk does not need to rush for more output to earn more.
- Any reasonable answer

2@, max: 2 marks

(Mark the **FIRST** point only)

From the viewpoint of the clerk:

- The income is more stable.
- Any reasonable answer

2@, max: 2 marks

(Mark the **FIRST** point only)

- (c) (i) No. Increasing the amount of the compensation plan would increase the value of relocating the bee farm without affecting the opportunity cost of the relocation. (2 marks)
- (ii) Yes. The increase in the wage rate of logistics workers will increase the monetary cost of relocating the bee farm. (2 marks)
7. (a) An example of a variable cost:
- wages of riders
 - cost of purchasing takeaway bags
 - any reasonable answer
- (Mark the **FIRST** point only)
- (b) Reasons:
- Different platforms provide heterogeneous services, e.g. different food provided by different partner restaurants.
 - Market information is imperfect, e.g. information on partner restaurants and service quality.
 - The platforms engage in price competition and non-price competition, e.g. advertising.
 - The platforms are price searchers, e.g. they charge different delivery fees.
 - Any reasonable answers
- (Mark the **FIRST TWO** points only)
- (c) After waiving the delivery fee, the relative price of lower-priced orders (in terms of higher-priced orders) will decrease. According to the law of demand, people will consume more lower-priced orders relative to higher-priced orders. The proportion of lower-priced orders to the higher-priced orders will increase. (4 marks)
8. (a) The pollutants emitted by factories pollute the air and worsen the air quality, which imposes costs on others (such as negative effects on the health of others). However, people are not compensated for those costs by the factories, so external cost exists. The marginal social cost is greater than the marginal social benefit. The amount of pollutants emitted by the factories is higher than the efficient output level. (3 marks)

1@, max: 1 mark

2@, max: 4 marks

(b) Verbal elaboration:

Imposing the additional sewage charge will increase the marginal cost of the industrial firms.

(1 mark)

The supply of the industrial firms will decrease.

(1 mark)

Producer surplus will decrease.

(1 mark)

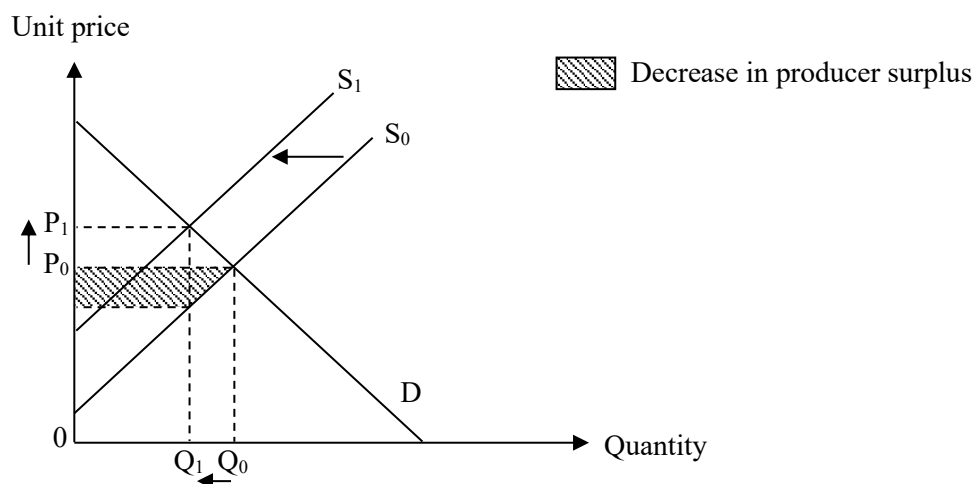
Illustrate in the diagram:

- leftward shift of the supply curve

(1 mark)

- correct position of the decrease in producer surplus

(1 mark)



- (c) The quantity of fresh air available is not sufficient to satisfy all human wants and people prefer more of it. / She is willing to pay for the fresh air.

(2 marks)

(d) Marks awarded for effective communication (EC: max 2 marks)

Marks	Performance
2	• Arguments supported with the source/data and appropriate economic theories • Relevant materials presented • Well-organised and coherent answers presented without repetition of ideas • Ideas clearly and fluently expressed with appropriate use of language/words/terms/symbols
1	• Arguments presented with some support of the source/data and economic theories • Some irrelevant materials presented • Answers presented in a less organised way with some repetition of ideas • A clear message conveyed with some inappropriate use of language/words/terms/symbols
0	• Arguments presented with no support of the source/data and economic theories • Materials unrelated to the gist of the question presented • Inconsistent arguments presented • Limited ideas expressed with inappropriate use of words/terms/symbols

The maximum marks for content are 9 marks. Answers may include the following:

Private car drivers

- Electric cars and diesel cars are substitutes. Providing a tax concession for first-time buyers of electric cars will reduce the cost of purchasing electric cars. Some diesel private car drivers may switch to electric cars. They would prefer proposal I.
- For drivers who are used to using private cars, they do not directly benefit from lower public transport fares, so they would not prefer proposal II.
- Lowering public transport fares may reduce people's demand for private cars, and the price of private cars may fall. In addition, when people reduce the use of private cars, the problem of traffic congestion will be alleviated. The time cost of private car drivers using the roads will decrease. Private car drivers would prefer proposal II.

Low-income earners

- High-income people usually use private cars while low-income people usually use public transport. Providing a tax concession for first-time buyers of electric cars mainly benefits high-income people and has little effect on low-income people. This means that providing a tax concession for first-time buyers of electric cars will worsen income inequality. Low-income people would not prefer proposal I.

- Low-income people usually use public transport to travel. Lowering public transport fares will increase their consumer surplus. If low-income people's demand for public transport is inelastic, lowering public transport fares will reduce their total expenditure on public transport, so they would prefer proposal II.
- Lowering public transport fares will encourage people to use more public transport. If there is excess demand for public transport, lowering the fares will increase the excess demand and people may have to wait for a longer time to take public transport, which will increase the time cost of using public transport for low-income people. If the increase in time cost is greater than the decrease in money cost, the full cost of using public transport will increase. Low-income people would not prefer proposal II.

Environmentalists who are concerned about air pollution

- Electric cars and diesel cars are substitutes. Providing a tax concession for first-time buyers of electric cars will lead to a decrease in the demand for diesel cars. The number of diesel cars on roads will decrease and the pollutants emitted by diesel cars will therefore decrease. Environmentalists who are concerned about air pollution would prefer proposal I.
- When more drivers switch to public transport due to the reduction in public transport fares, the number of private cars on roads will decrease and pollutants will decrease. Environmentalists who are concerned about air pollution would prefer proposal II.